

Criterion E

On the basis of the evaluation interview that was conducted with the client post multiple weeks of client testing the following was evidenced:

Success Criterion		Met?	Client Feedback	Client score
1	Coaches should be able to select a video and view it in the webpage with the ability to slow the speed for scrutiny of the video prior to submission.		"I was able to choose a video and it would pop up on the screen" "being able to make the video full screen and slow its playing speed made it much easier for me to watch the delivery and analyse what is going on before I submit it"	9/10
2	Coaches should be able to upload a video up to 5 seconds in length, of each delivery bowled by the student to the application.		"I never really faced an issue in uploading videos to the application" "I think that the 5-second limit was good...since it made me crop the video to only contain my student's bowling" "even on submitting a file that is too big, I was reminded to go back and crop the video"	10/10
3	The application should be able to produce an output video that clearly shows the trajectory path of the ball using a visible white line.		"the quality of ball tracking was actually surprisingly high" "when watching a plain video it is difficult to watch the trajectory and see changes and this solution made this easy"	9/10
4	The application should be able to determine how many meters the ball bounces away from the stumps of the striker.		"This feature was quite functional and was pretty good." "might not be required but the red dot in the video provided good visual aid"	8/10
5	The application should be able to determine how many meters the ball bounces away from the stumps of the striker.		"This meter calculation was very helpful" "the exact calculation of meters assisted me to have an objective numeric point of analysis of the length" "I would still give it a 95% success rate"	8/10

6	The application should be able to produce a pitch map graphic on which the bounce spot is shown by a dot drawn at the bounce position of the specific delivery in the submitted video.		"In the Pitch graphics the balls were lined up with the centre of the stumps meaning that they are only finding the length and not the line" Partially unfulfilled success criteria	5/10
7	The application should be able to produce a pitch map that displays the bounce spot of all deliveries that have been uploaded to the application to date.		"the master pitchmap was actually really helpful, in terms of providing a point of comparison" "allows me to compare analysis to the deliveries from the ball previously"	8/10

Recommendations

Client Recommendations

My client was happy with the final product and agreed that it met his needs as evidenced by the evaluation interview. Nonetheless, the below areas were identified by the clients to improve upon:

1. **Easier Method of importing videos:**
 - a. Client felt that it was tedious to import videos from the phone to the PC to use in the Flask application. A possible alternative would be to use a third-party application for Android and iOS devices that facilitate easier video transfer.
2. **Determine the speed of bowling:**
 - a. Client felt that his coaching would benefit greatly if the speed of delivery was calculated. This can be done by using the length of the recording and the constant distance of the pitch the ball has to travel.
3. **Calculation of line:**
 - a. This can be implemented using OpenCV2s functions for perspective distortion.

Developer Recommendations

1. **Add a database** to store the information on newly added features. For example, if speed analysis is added then a database can be used to store more information on each delivery.
2. **Train a more robust Yolov5 model** that can perform more accurate object detection even at a distance so that the application can be used for game scenarios.

Word count: 497